

MOHONA YESMIN

[Portfolio](#) | [GitHub](#) | [LinkedIn](#) | myesmin415@gmail.com | Bronx, NY

EDUCATION

- Hunter College**, New York, NY May 2028
Bachelor of Arts, Major: Computer Science, Minor: Math and Statistics
- Wesleyan University**, Middletown, CT May 2023
Bachelor of Arts, Major: Neuroscience and Behavior, Minor: Chemistry and Film, Extensive Coursework in Data Analytics

TECHNICAL SKILLS & CERTIFICATION

AI & Machine Learning: Generative AI (LLMs, Agentic Frameworks, MCP, API Integration), Claude Code, Google AI Studio, Graph Neural Networks (PyTorch Geometric), Statistical Modeling, NLP, Neural Networks, Predictive Modeling, Feature Engineering
Data Engineering & Visualization: Web Scraping, ETL Pipelines, Tableau, Matplotlib, Seaborn, Plotly, Power BI
Languages: Python, C++, SQL **Frameworks:** pandas, LangChain, TensorFlow, PyTorch, Scikit-learn, FastAPI, PySpark, Streamlit
DevOps & Productivity: Git, GitHub, Supabase, Vercel, Docker, Jira, Power Automate, Microsoft Office Suite, Google Workspace
Certifications: Hugging Face AI Agents, McKinsey Forward Program

PROJECTS

- Kubeflow (CNCF Open-Source ML Platform)**, *Org Member & Contributor* January 2026 – Present
- Shipped merged PRs migrating Python clients off deprecated APIs to v1 per KEP-0004 across kubeflow/hub and kubeflow/sdk, regenerating OpenAPI clients and validating 270+ unit/E2E tests with pytest, mypy, ruff
 - Designed and shipped a decoupled upload_artifact utility (Python, Pydantic) supporting S3 and OCI object storage, isolating artifact uploads from registry logic per a community feature request
 - Building a Go client and backend proxy layer routing UI traffic to v1 Model Registry, Model Catalog, and MCP Catalog APIs; co-authored the SDK 0.4.0 release blog post and became a Kubeflow org member on the strength of sustained contributions
- Real-Estate Valuation & Risk**, [Dashboard](#) | [GitHub](#) 2026
- Built an automated valuation model on 2,412 sales (LightGBM, DuckDB SQL, conformal prediction) with out-of-time validation, quarterly-recalibrated price ranges, drift monitoring and containerized FastAPI service; reached 5.5% median error, 77% of homes within $\pm 10\%$, showed valuation uncertainty adds 85% to expected losses, and stress-tested a loan book under a 30% price drop
- Climate Risk & Emissions**, [Dashboard](#) | [GitHub](#) 2026
- Built a Python forecasting and scenario pipeline across 190+ economies from six public sources, with an interactive Plotly dashboard; found a 23 $\times$ US and Bangladesh per-person emissions gap and that even fast decarbonization overshoots the 2 °C carbon budget

EXPERIENCE

- DAIR Lab**, *Research Assistant*, New York, NY August 2026 – Present
- Scaled a multimodal GNN pipeline (PyTorch Geometric GAT/GCN encoders, ESM protein embeddings, GUDHI persistent homology) for toxin ion-channel classification to a 200-protein benchmark at 87.5% test accuracy; building domain-adversarial training with a gradient reversal layer to close a ~30-point gap between random-split (~86%) and held-out-taxon (~55%) accuracy
- EN-POWER GROUP**, *Energy Data Manager I* | *Energy Data Intern*, New York, NY June 2025 – Present
- Developed and deployed statistical models to predict building emissions trajectories across 200+ building portfolios, delivering interpretable, actionable LL97 compliance recommendations that drove business value for 50+ clients.
 - Engineered large-scale data pipelines to scrape, transform, and aggregate multi-source energy datasets, building client-facing dashboards and supporting LL97 compliance filings and payment processing for 900+ properties in coordination with DOB systems
 - Automated client communications, document workflows, and file management across 1,000+ clients using Python-based tools, reducing manual processing time by more than 50%
- Aarohaa.AI**, *Applied AI Intern*, New York, NY May 2025 – June 2025
- Designed and deployed a JSON parsing agent to normalize LLM outputs across multiple data formats, ensuring consistent inter-agent communication and scalable multi-agent orchestration using RAG and agentic AI pipelines
 - Implemented FastAPI endpoints to integrate AI agents with live dashboards, enabling real-time data processing, decision support, and system transparency in production environments
 - Extracted, transformed, and validated backend SQL datasets by engineering new data columns and persistence logic to ensure accurate, reliable data storage across agent operations
- DRUM**, *Campaign Strategist & Data Lead* | *Organizing Fellow*, Bronx, NY September 2023 – January 2026
- Designed and analyzed 100+ community surveys and maintained a 500+ person membership database, translating multi-source data into visualizations and clear reports that guided resource allocation for non-technical stakeholders and decision-makers
- General Assembly**, *Data Science Immersive Bootcamp fellow*, New York, NY September 2023 – December 2023
- Completed **500+ hours** of training in statistical modeling, machine learning algorithms, and advanced mathematics, with applications in fintech & AI using Python and SQL
 - Built and deployed ML models for regression, classification, NLP, LLMs and anomaly detection across financial risk prediction, market trend forecasting and portfolio optimization, with feature engineering, validation and Streamlit dashboards